

CSE 331

AUTOMATA AND COMPUTABILITY

Lecture 3

Spring 2026

Today's learning goals

Sipser Ch 1.1, 1.2

- Review what it means for a **set** to be closed under an **operation**.
- Define the regular operations on languages
- Prove closure properties of the class of regular languages

Building DFA



Remember

States are our only (computer) memory.

Design and pick states with specific roles / tasks in mind.

"Have not seen any of desired pattern yet"

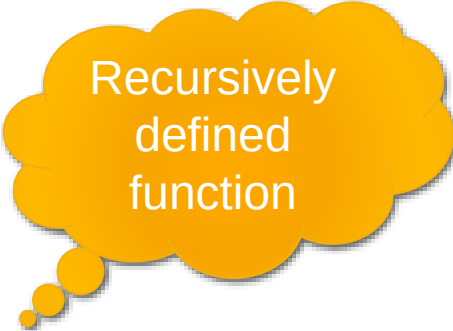
"Trap state"

A useful (optional) bit of terminology

When is a string accepted by a DFA?

Computation of M on w: *where do we land when start at q_0 and read each symbol of w one-at-a time?*

$$\delta^*(q, w) =$$



Recursively
defined
function

Regular languages

Sipser p. 35 Def 1.5

- For DFA M over the alphabet Σ
 - For each string w over Σ , M either accepts w or rejects w
 - The **language recognized by M** is the set of strings M accepts
 - The **language of M** is the set of strings M accepts
 - **$L(M)$** = $\{ w \mid w \text{ is a string over } \Sigma \text{ and } M \text{ accepts } w \}$

Definition: A language is **regular** iff there is some finite automaton that recognizes **exactly** it.

Regular languages: bounds?

Is **every** finite language regular?

- A. No: some finite languages are regular, and some are not.
- B. No: there are no finite regular languages.
- C. Yes: every finite language is regular.
- D. I don't know.

Building DFA



New strategy

Express L in terms of simpler languages – use them as building blocks.

Example

$L = \{ w \mid w \text{ does not contain the substring baba} \}$
= the complement of the set
 $\{w \mid w \text{ contains the substring baba}\}$

Building DFA



DFA recognizing

$\{w \mid w \text{ contains the substring baba}\}$

DFA recognizing

$\{w \mid w \text{ doesn't contain the substring baba}\}$

Complementation

Theorem: If A is a regular language over $\{0,1\}^*$, then so is its complement $\bar{A}$

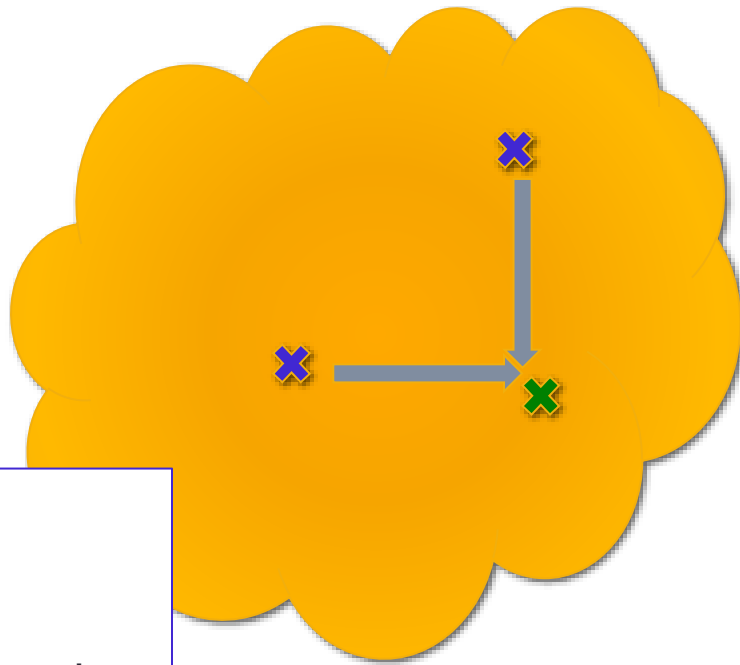
aka "the class of regular languages is closed under complementation"

Closure of ... under ...

- $\mathbb{Z}$ under addition.
- Set of even ints under multiplication.
- $\{0\}^*$ under concatenation.

Which of these is true?

- A. The set of odd integers is closed under addition.
- B. The set of positive integers is closed under subtraction.
- C. The set of rational numbers is closed under multiplication.
- D. The set of real numbers is closed under division.
- E. I don't know.



Complementation

Theorem: If A is a regular language over $\{0,1\}^*$, then so is its complement $\bar{A}$

aka "the class of regular languages is closed under complementation"

Complementation

Theorem: If A is a regular language over $\{0,1\}^*$, then so is its complement $\bar{A}$

aka "the class of regular languages is closed under complementation"

Proof: Let A be a regular language.

Then there is a DFA $M=(Q,\Sigma,\delta,q_0,F)$ such that $L(M) = A$.

We want to build a DFA M' whose language is $\bar{A}$. Define

$M' =$

?

Claim of Correctness $L(M') = \bar{A}$

Proof of claim...



Why closure proofs?

- General technique of proving a new language is regular
- Stretch the power of the model
- Puzzle!

The regular operations

Sipser Def 1.23 p. 44

For A, B languages over same alphabet, define:

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

$$A \circ B = \{xy \mid x \in A \text{ and } y \in B\}$$

$$A^* = \{x_1x_2 \dots x_k \mid k \geq 0 \text{ and each } x_i \in A\}$$



These are operations on sets!

Union

Sipser Theorem 1.25 p. 45

Theorem: The class of regular languages over fixed alphabet Σ is closed under the union operation.

Proof:

What are we proving here?

- A. For any set A , if A is regular then so is $A \cup A$.
- B. For any sets A and B , if $A \cup B$ is regular, then so is A .
- C. For two DFAs M_1 and M_2 , $M_1 \cup M_2$ is regular.
- D. None of the above.
- E. I don't know.

Union

Sipser Theorem 1.25 p. 45

Theorem: The class of regular languages over fixed alphabet Σ is closed under the union operation.

Proof: Let A_1, A_2 be any two regular languages over Σ .

WTS that $A_1 \cup A_2$ is regular.

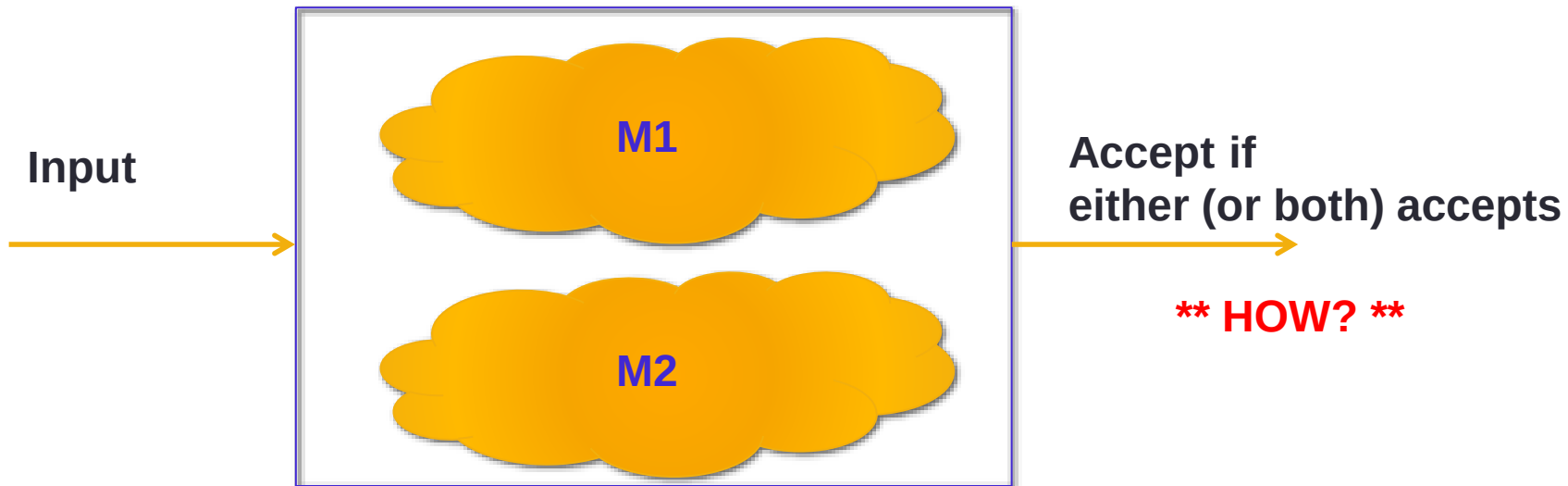
Goal: build a DFA that recognizes $A_1 \cup A_2$.

Union

Sipser Theorem 1.25 p. 45

Goal: build a DFA that recognizes $A1 \cup A2$.

Strategy: use DFAs that recognize each of $A1, A2$.



Union

Sipser Theorem 1.25 p. 45

Theorem: The class of regular languages over fixed alphabet Σ is closed under the union operation.

Proof: Let A_1, A_2 be any two regular languages over Σ .

Given $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ such that $L(M_1) = A_1$ and

$M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ such that $L(M_2) = A_2$

WTS that $A_1 \cup A_2$ is regular.

Define $M = (?, \Sigma, \delta, ?, ?)$

Union

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Proof: Let A_1, A_2 be any two regular languages over Σ .

Given $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ such that $L(M_1) = A_1$ and

$M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ such that $L(M_2) = A_2$ and

WTS that $A_1 \cup A_2$ is regular.

Define $M = (Q_1 \times Q_2, \Sigma, \delta, ?, ?)$. . .

Idea: run in parallel

Union

Sipser Theorem 1.25 p. 45

Theorem: The class of regular languages over fixed alphabet Σ is closed under the union operation.

Proof: Let A_1, A_2 be any two regular languages over Σ .

Given $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ such that $L(M_1) = A_1$ and $M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ such that $L(M_2) = A_2$, what should be the initial state of M ?

$M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ such that

WTS that $A_1 \cup A_2$ is regular.

Define $M = (Q_1 \times Q_2, \Sigma, \delta, ?, ?)$

What should be the initial state of M ?

- A. q_0
- B. q_1
- C. q_2
- D. (q_1, q_2)
- E. I don't know.

Union

Sipser Theorem 1.25 p. 45

Theorem: The class of regular languages over fixed alphabet Σ is closed under the union operation.

Proof: Let A_1, A_2 be any two

Given $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$:

$M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ such th

WTS that $A_1 \cup A_2$ is regular.

Define $M = (Q_1 \times Q_2, \Sigma, \delta, ?, ?)$

When r is a state in M_1 , s is a state in M_2 ,
and x is in Σ , then $\delta((r,s), x) =$

- A. (r,s)
- B. $(\delta(r,x), \delta(s,x))$
- C. $(\delta_1(r,x), s)$
- D. $(\delta_1(r,x), \delta_2(s,x))$
- E. I don't know.

Union

Sipser Theorem 1.25 p. 45

Theorem: The class of regular languages over fixed alphabet Σ is closed under the union operation.

Proof: Let A_1, A_2 be any two

Given $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$:

$M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ such th

WTS that $A_1 \cup A_2$ is regular.

Define $M = (Q_1 \times Q_2, \Sigma, \delta, ?, ?)$

The set of accepting states for M is

- A. $F_1 \times F_2$
- B. $\{ (r,s) \mid r \text{ is in } F_1 \text{ and } s \text{ is in } F_2 \}$
- C. $\{ (r,s) \mid r \text{ is in } F_1 \text{ or } s \text{ is in } F_2 \}$
- D. $F_1 \cup F_2$
- E. I don't know.

Union

Sipser Theorem 1.25 p. 45

Proof: Let A_1, A_2 be any two regular languages over Σ .
Given $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ such that $L(M_1) = A_1$ and
 $M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ such that $L(M_2) = A_2$.

WTS that $A_1 \cup A_2$ is regular. Define

$M = (Q_1 \times Q_2, \Sigma, \delta, (q_1, q_2), \{(r, s) \text{ in } Q_1 \times Q_2 \mid r \text{ in } F_1 \text{ or } s \text{ in } F_2\})$
with $\delta((r, s), x) = (\delta_1(r, x), \delta_2(s, x))$ for each $(r, s) \text{ in } Q_1 \times Q_2$
and $x \text{ in } \Sigma$.

Why does $L(M) = A_1 \cup A_2$?

Intersection

- *How would you prove that the class of regular languages is closed under intersection?*
- *Can you think of **more than one** proof strategy?*

$$A \cap B = \{ x \mid x \text{ in } A \text{ and } x \text{ in } B \}$$

Payoff

$\{ w \mid w \text{ contains neither the substrings aba nor baab} \}$

Is this a regular set?

Payoff

$\{ w \mid w \text{ contains neither the substrings aba nor baab} \}$

Is this a regular set?

$A = \{ w \mid w \text{ contains aba as a substring} \}$

$B = \{ w \mid w \text{ contains baab as a substring} \}$

$$\bar{A} \cap \bar{B} = \overline{A \cup B}$$

General proof structure/strategy

Theorem: For any L over Σ , if L is regular then [the result of some operation on L] is also regular.

Proof:

Given name variables for sets, DFAs assumed to exist.

WTS state goal and outline plan.

Construction using objects previously defined + new tools working towards goal. Give formal definition and explain.

Correctness prove that construction works.

Conclusion recap what you've proved.

The regular operations

Sipser Def 1.23 p. 44

For A, B languages over same alphabet, define:

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\} \quad \checkmark$$

$$A \circ B = \{xy \mid x \in A \text{ and } y \in B\}$$

$$A^* = \{x_1x_2 \dots x_k \mid k \geq 0 \text{ and each } x_i \in A\}$$



How can we prove that the concatenation of two regular languages is a regular language?